

Soroush Saririan, EIT

saririan.s.eng@gmail.com | (518) 847-9639 | linkedin.com/in/soroushsaririan | soroushsaririan.github.io

EDUCATION

M.S. Applied Data Science | University of Florida, Gainesville, FL **Aug 2024 – May 2026**
GPA: 3.97

B.E. Mechanical Engineering | Stony Brook University, Stony Brook, NY **Feb 2021 – May 2024**
GPA: 3.70 | *Magna Cum Laude, Dean's List*

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Rust, MATLAB

ML and AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, SciPy, LangGraph, LangChain, LlamaIndex, Hugging Face, LLMs, VLMs, RAG, XGBoost, CatBoost, Matplotlib, Seaborn

DevOps and Tools: Docker, Kubernetes, Git/GitHub, Linux, AWS, FastAPI, ChromaDB, FAISS, PostgreSQL, NVIDIA NIM, Streamlit, Arduino, Tableau, Excel, Power BI

Domain: Multi-Agent AI Systems, Computer Vision, Clinical Decision Support, Time-Series Analysis, sEMG Signal Processing, Static Code Analysis, Statistical Modeling, Design of Experiments, A/B Testing

WORK EXPERIENCE

AI/ML Engineer (Graduate Co-op) **Aug 2025 – Present**
Raytheon, Largo, FL

- Engineered an LLM-assisted static analysis pipeline using DARPA AI Cyber Grand Challenge technology to translate legacy Ada codebases into memory-safe C++ and Rust, eliminating 912 engineering-hours of manual refactoring effort
- Integrated multiple Cyber Reasoning Systems (CRSs) with LLM-driven source analysis to automatically detect vulnerabilities and generate patches, delivering automated security reports to engineering review boards
- Designed evaluation harnesses to benchmark patch correctness across Ada-to-Rust translation tasks, achieving 80% recall and enabling iterative LLM prompt refinement that improved patch acceptance rates

Sports Medicine Data Researcher **Sep 2025 – Present**
University of Florida, Gainesville, FL

- Engineered Python ingestion pipelines for Shimmer3 sEMG streams using bandpass filtering, RMS windowing, and onset detection to extract physiological features from raw time-series signals
- Modeled muscle activation patterns in stroke patient gait cycles via statistical feature extraction, producing quantitative metrics that directly informed clinical rehabilitation assessment
- Iterated on windowing configurations and validated parameters against clinical benchmarks, improving onset detection reliability and feature consistency across patient recording sessions

Undergraduate Researcher **Feb 2023 – May 2024**
Stony Brook University, Stony Brook, NY

- Designed compressive testing protocols across 100+ CNT-reinforced epoxy nanocomposite samples, varying reinforcement geometry to isolate the mechanical contribution of straight vs. helical carbon nanotubes
- Applied statistical regression to identify a 15–20% strength gain from optimized CNT geometry; published as lead author in *Polymer Composites* (doi.org/10.1002/pc.29076) and awarded the Richard S. Lee Research Excellence Award

PROJECTS

Automated Skin Allergen Imaging Pipeline **Aug 2023 – Present**
Independent Research

- Built an OpenCV pipeline with an ensemble classifier (Random Forest, SVM, CNN) trained on 500+ clinical photographs to automate skin prick test wheal detection and measurement
- Achieved greater than 90% clinician-score correlation, 95% wheal detection accuracy, and 85% reduction in diagnosis time vs. manual scoring

Wearable Sensor Data Acquisition System **2024**
Independent Project

- Architected an Arduino and MPU6050 embedded logging system capturing 6-DOF kinematic time-series data
- Developed Python serial extraction and CSV preprocessing routines for downstream gait feature analysis